Name _____ _____ Baggett/Fall2026/207/Quiz3

first name last name

Score / 20

No Calculators.

Questions 1 and 2. Circle **True** or **False**. Then, in **one sentence**, say why — if the statement is false, the fastest way is usually to give an example that breaks it. **1 point for the circle, 2 for the reason.**

1. (**3** pts) A function can have two different horizontal asymptotes.

True **False**

Why? _____ /3

2. (**3** pts) If $g(-2) = -5$ and $g(4) = 7$, then $g(c) = 0$ for some c in $(-2, 4)$.

True **False**

Why? _____ /3

3. (**4** pts) Evaluate the limit. **Show the step where the sign of x matters.**

$$\lim_{x \to -\infty} \frac{\sqrt{4x^2 + 3}}{x + 5}$$

_____ /4

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4. (**5** pts) Let $f(x) = x^2 + 2x$.

(a) Find $f'(3)$ **from the definition**. Keep lim on every line until you actually take it.

(b) Write the equation of the tangent line to $y = f(x)$ at $x = 3$.

/5

5. (**3** pts) Show that the equation $x^3 + x = 1$ has a solution between 0 and 1. **Name the theorem you use, and check its hypothesis.**

/3

6. (**2** pts) Write the equation of the horizontal asymptote of $f(x) = \dfrac{3x^2 - 1}{x^2 + 4}$.

/2

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